

CUSTOMER SHOPPING BEHAVIOR ANALYSIS REPORT

1. Project Overview:

The project analyses customer buying pattern data across 3900 purchases across different product categories. The objective of the project is to identify patterns and find insights in the purchasing pattern of product based on categories, age group, subscription, gender, etc and help strategizing business decision.

2. Dataset Summary

The dataset has 3900 rows and 18 columns the column names are : 'customer_id', 'age', 'gender', 'item_purchased', 'category', 'purchase_amount', 'location', 'size', 'color', 'season', 'review_rating', 'subscription_status', 'shipping_type', 'discount_applied', 'promo_code_used', 'previous_purchases', 'payment_method', 'frequency_of_purchases'

3. Analysis And Data Cleaning Using Python

We started the data analysis in python :

- **Data Loading:** Importing data in python
- **Initial Data Exploration:** Used function like `df.info()` to check structure of the data and `df.describe()` to reveal the statistical insights from the data.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                       3900 non-null   object
4   Category                             3900 non-null   object
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                 3900 non-null   object
8   Color                                3900 non-null   object
9   Season                               3900 non-null   object
10  Review Rating                        3863 non-null   float64
11  Subscription Status                  3900 non-null   object
12  Shipping Type                       3900 non-null   object
13  Discount Applied                    3900 non-null   object
14  Promo Code Used                     3900 non-null   object
15  Previous Purchases                  3900 non-null   int64
16  Payment Method                      3900 non-null   object
17  Frequency of Purchases               3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- **Manipulating Column Names:** Updated all column names to lower case and replaced spaces with underscore in a snake case form.
- **Finding null values in Columns:** Searched for null values inside the column. It was found that 37 data of column review rating is missing.

```

Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    37
Subscription Status  0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method   0
Frequency of Purchases  0
dtype: int64

```

- **Replacing Null Values:** Replaced null values of review rating column with the median value of the respective category.
- **Feature Addition:**
 - Distributed ages among 4 categories young adult, adult, middle aged, senior
 - Specified days from [frequency_of_purchase](#) column
- **Removed Redundant Columns:** [promo_code_used](#) constituted similar data to [discount_applied](#) column and was dropped
- **Database Integration:** Connected python script to PostgreSQL **database** and imported the cleaned data frame into the SQL database for further analysis

4. Answering Business Question Using SQL

- Total revenue generated by male v/s female :

	gender text	revenue numeric
1	Female	75191
2	Male	157890

- Customers used a discount but spent more than the average purchase amount

	customer_id bigint	purchase_amount bigint
27	65	83
28	67	94
29	70	70
30	74	85
31	76	85

Total rows: 839 Query complete 00:00:00.207

- Top best rated products according to customer:

	item_purchased text	Average Ratings numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

- Average purchase amount between standard and express shipping

	shipping_type text	average payment numeric
1	Standard	58.46
2	Express	60.48

- Average spent and total revenue between subscriber and non-subscriber

	subscription_status text	total_customers bigint	Total amount received numeric	average amount numeric
1	No	2847	170436.00	59.87
2	Yes	1053	62645.00	59.49

- Products have the highest percentage of purchases with discounts applied

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

- 3 topmost purchased product within each category

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

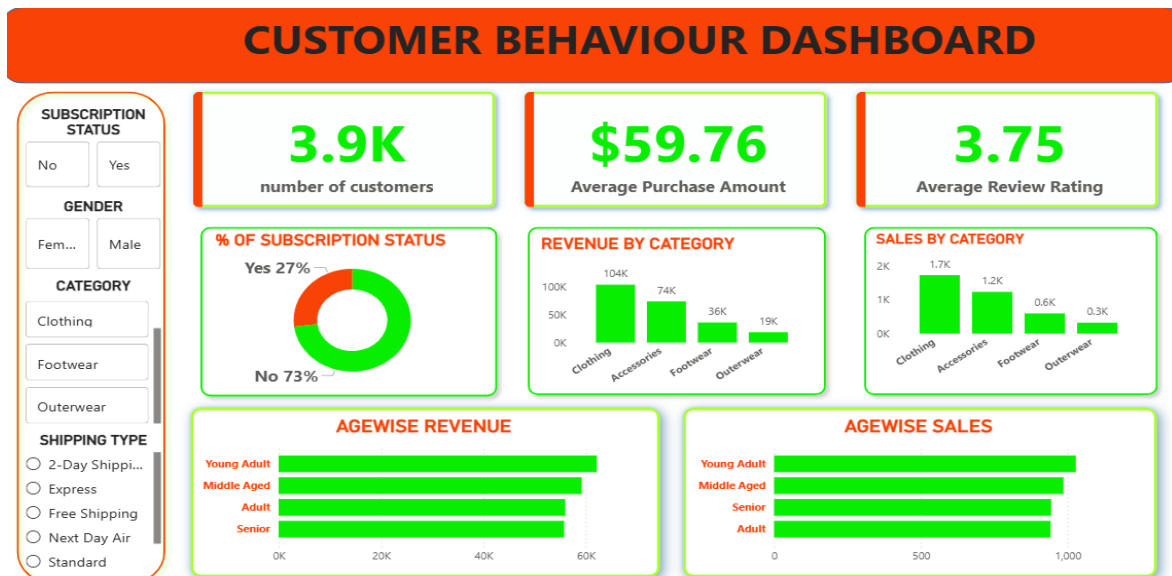
- Are customer who are repeat buyers (more than 5 previous purchases) also like to subscribe

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

- The revenue contribution of each age group

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle Aged	59197
3	Adult	55978
4	Senior	55763

5. Data Visualization using Power Bi



6. Business Recommendation

- **Target Marketing Towards Young People:** Market or positioned the products to the newer generation.
- **Provide Discount On Lowest Selling Product:** Provide heavy discount in outerwear products while keeping the margin to increase the sales of Outerwear.
- **Subscription:** Need to add exclusive subscription feature to increase the subscription revenue.
- **Increasing Profit Margin:** Sell the topmost purchased products in each category at a conveniently higher price to improve the profit margins.
- **Delivery Speed:** Reduce the deliver time to attract more customer